

Numerical Methods

BCA — Semester IV — 2023 — Answer Sheet
Subject Code: CACS252

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions [2x10=20]

2. Define error. Explain the Taxonomy Errors.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Estimate the value of $\sin \theta$ at $\theta = 45^\circ$ using Newton's backward difference formula from the following set of data.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Write an algorithm and program to calculate integration using Trapezoidal rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is the form of resultant matrix using Gauss-Jordan method? Solve the following system of equations using Gauss-Jordan Method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define ordinary differential equation. Use the fourth order Runge-Kutta method to estimate $y(0.4)$ of the equation: $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$ assuming that $h = 0.2$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Solve for the steady state temperatures in a rectangular plate of 8cm x 10cm, if one 10cm side is held at 50°C, and the other 10cm side is held at 30°C and other two sides are held at 10°C. Assume grids of size 2cm x 2cm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write a short note on (Any Two):
a) Partial Differential Equations
b) Linear Interpretation
c) Boundary value problems

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Write an algorithm and program to compute the root of nonlinear equation using Newton - Raphson method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. a) Fit a straight line to the following set of data using Least Square Regression. b) Apply the factorization method (any) to solve the equations: $2x + 3y + 7z = 42$, $x + 3y + z = 53$, $x + 4y + z = 7$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. a) Write and implement an algorithm to solve the system of linear equations using Gauss-Seidel method with suitable example. b) Write a program to solve the ordinary differential equations using Heun's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.